Final Report – Human-Centered AI

Prototype Design



CREATING MEANINGFUL EXPERIENCES

Applied Data Science & AI

Date

Breda University OF APPLIED SCIENCES

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Task 4.1: Create a Wireframe Prototype

Wireframe-Version B
Wireframe-Version B on GitHub
Wireframe-Version B Figma Link

My Design Choices (Using Human-Centered AI Principles)

When I designed ScanFresh, I wanted it to be easy, helpful, and respectful to users. I used human-centered AI design principles to make sure people stay in control, understand the AI, and feel supported while using the app.

1. Easy and Friendly Start
I made a clean splash and welcome screen with bright food images so the app feels fresh and welcoming. The login screen has simple options like Google, Facebook, and email, so anyone can start easily.
Principle: Accessibility & Inclusiveness

2. Clear and Simple Main Page
I gave users a clear choice between "Fruits" and "Vegetables" with big buttons. The options like "Start Scanning" and "Upload from Gallery" are easy to find and understand.
Principle: User Control

3. Helpful Scanning Process
When scanning, users see clear food photos and visual hints (like circles) showing where the AI is looking. This helps users understand what's happening.
Principle: Transparent AI

4. Respectful and Useful Feedback
After scanning, users are asked, "Was this helpful?" and can give star ratings. If something goes wrong, the app says, "Sorry for the confusion," and lets them try again. This shows that the app cares about user input and learns from it.
Principle: Feedback & Human-AI Collaboration

5. Easy-to-Use Scan History
Users can see what they scanned before with photos and names. This helps them track what they did and builds trust with the app.
Principle: Transparent AI

6. Simple Profile Page
I kept the profile page clean with just the name, image, and logout. It respects privacy and gives users control over their account.
Principle: User Control & Respect for Privacy

7. Final Summary:

I designed ScanFresh to be human-centered — the AI helps, but the user stays in control. I used principles like transparent AI, feedback, user control, and simplicity to make sure the app feels easy, respectful, and helpful.

Task 4.2: Think-Aloud Study

A. Method

For my think-aloud study, I wanted to understand how easy it is for users to use the Scan Fresh app and if they face any problems while using it. The goal was to see if the design is clear and simple to use. The participants in this study were 3 of my peers. They helped me gather feedback on the app's usability and identify any areas for improvement.

Materials Used:

- Figma Wireframes: I used a prototype of the app created in Figma.
- Teams: The study was done using Microsoft Teams for video calls.
- Screen Recording: I recorded the users' screens to see how they interacted with the app.
- Audio Recording: I also recorded the users talking to understand their thoughts while using the app.

User Tasks:

- Open the app.
- Log in or sign up.
- Try to scan an image to see the freshness results.
- Navigate to the scan history.
- Upload an image and see the results.
- Check the scan history to see past results.

Interview Questions:

- What did you like about the app?
- Did you understand how to take or upload a photo?
- Was the log-in screen clear to you?
- What was frustrating or confusing?
- Any suggestions to improve the app?

B. Data

Transcript exported peer1.txt
Transcript exported peer 2.txt
Transcript exported peer 3.txt

C. Results

To analyse the results of the think-aloud study, I used thematic analysis, which involves identifying common themes and patterns across participant feedback. After transcribing each session, I read through the responses and coded key actions and comments, such as whether the user knew what to do, was confused, or gave suggestions. I then grouped these codes into larger themes that appeared across multiple participants.

Theme	Description	Participant(s)	Design Principle
Easy and Simple Interface	Users found the app clean, simple, and easy to navigate.	P1, P3, P2	User Control & Simplicity
Lack of Scanning Feedback	Users didn't fully understand what was happening during scanning.	P3, P2	Transparent AI & Feedback
Confusing Login Options	One user was unsure which login method to choose; needed more guidance.	P2	Accessibility & User Guidance
Need for First-Time Help	Users suggested adding a short tutorial or help screen at the beginning.	P1, P2	Inclusiveness & Onboarding
Want More Info on Results	User wanted more explanation about result meanings like "Almost Fresh" or percentage scores.	P2	AI Transparency & Trust

D. Discussion

How I Used the Feedback to Improve the App

The study showed that the app is easy to use, which is good. But some users were confused during scanning, so I added a "Scanning..." message to make it more clear. One person didn't understand the login buttons, so I added text under each button, like "Login with Facebook" next to the Facebook icon. Two users asked for help for first-time use, but I didn't change anything because I will make a demo video at the end to explain how to use the app. One user wanted more info about the results, so I added an info button to explain things like "80% Fresh" (green for good) or "40% Spoiled" (red for bad). These changes make the app easier and better for everyone.

Task 4.3: A/B Testing

A. Method

Describe your plan for the A/B test. Include versions A and B and elaborate upon the difference and the survey questions.

Goal: I want to see if adding a feedback question after a scan makes the app better for users.

- Test Versions:

Version A :

- Description: This version only has a button to start a new scan. After the scan, the user moves to the next step without any feedback option.
- Key Feature: "Start New Scan" button.

Version B:

- Description: This version has a feedback question that asks, "Was this helpful?" after the scan. The user can select "Yes" or "No."

- Key Feature: "Was this helpful?" question after each scan.

Difference Between Versions A and B:

- Version A: Users can scan and start a new scan without giving any feedback.
- Version B: After scanning, users can answer "Yes" or "No" to the "Was this helpful?" question.

User Task Flow (6 minutes):

Introduction (30 seconds):
- Explaining to the users what they need to do in the app.
- ask them to scan a fruit.

Testing (4 minutes):
- Version A:They will click "Start New Scan" and complete a scan.
- Version B: After the scan, they will answer "Yes" or "No" to the "Was this helpful?" question.

Survey (2 minutes):
- After trying both versions, I will ask them on survey the following questions:

Survey Questions:
1. I understood what I could use the app for
2. I found the application intuitive to use.
3. I thought the application was useful.
4. I enjoyed using the application.
5. Was the scanning process fast and easy to use?

Hypothesis:
- Version B : will make users feel better about the app because it asks for feedback.
- Version A : is faster but doesn't allow users to give feedback

Conclusion of the Test Plan:
The goal of this A/B test is to assess user preferences for the two versions of the app interface. I am aiming to see if the feedback mechanism in Version B improves user satisfaction or if Version A is more effective in driving user actions.

B. Data

Link to the cleaned data Version_A.csv
Link to the cleaned data Version_B(1-25).csv

C. Results

Link to the T-test notebook On GitHub

D. Discussion

What I Learned and How I Validated My Design

After testing both versions of the app, I saw that Version B (with the feedback question) got higher scores from users. But when I did the t-test and bootstrapped test, the results were not strong enough to prove that Version B is clearly better. This might be because only 13 people tested Version A, while 27 tested Version B so the groups were not even. Also, the sample sizes were small overall, and the data did not follow a normal distribution, which can make the results less reliable or harder to detect a real difference.
Even though the results were not statistically strong, the feedback showed that people liked Version B more. They liked being asked, "Was this helpful?" after scanning. It made the app feel more personal and respectful.
Because of that, I decided to keep the feedback question in my app. I think it helps users feel like their opinion matters. This follows my goal of making the app more human-centered — where people are part of the process, not just using a tool.
Next time, I will try to test with more users to get even better results. But for now, this test helped me make a small but meaningful improvement to the app.

Task 4.4: Create a Final Demo Video

❖ My Final Demo Video.mp4 On GitHub

References

Figma (n.d.). *Figma: Interface Design Tool*. Retrieved from https://www.figma.com
Microsoft. (n.d.). *Microsoft Forms*. Retrieved from https://forms.microsoft.com
OpenAI. (n.d.). *Introducing GPT-3*. Retrieved from https://openai.com/blog/openai-api



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Facility

Mgr. Hopmansstraat 2
4817 JS Breda

P.O. Box 3917
4800 DX Breda
The Netherlands

PHONE
+31 76 533 22 03
E-MAIL
communications@buas.nl
WEBSITE
www.BUas.nl

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